EPIGRAPHY AND THE HISTORICAL SCIENCES

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Latin Epigraphy and the IT Revolution

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Revolutions never seem to come soon enough for those who welcome them nor to end quickly enough for all caught up in them.[1] Certainly the IT revolution in epigraphy has seemed a long time in the making to those who more than thirty years ago began calling upon our international association to promote the creation of an electronic database of all the known Greek and Latin inscriptions of antiquity and thus to realize, in large measure, Niebuhr's dream of a comprehensive *Corpus Inscriptionum* of all the languages of the Roman Empire.[2] Niebuhr's proposal to the Berlin Academy of Sciences in 1815 languished for a decade before Boeckh got the first volume of a Greek corpus off the ground in 1825, and it was nearly another thirty years before Mommsen finally succeeded in 1853, against the opposition of Boeckh, in launching the great corpus of Latin inscriptions that he would guide and to a large extent drive for the next half century until his death in 1903. By that time most of the Latin inscriptions then known had been published in state-of-the-art editions that presented the texts with a typographical precision and elegance that have not since been surpassed. The undertaking was not only monumental but, as Professor Panciera noted in his introductory remarks,

[1] Since state-of-the-field reports of IT developments become outdated even sooner than other such reviews, their utility is less functional than commemorative. At best they provide a snapshot of a stage of evolution at a particular moment in time. The present contribution is no exception and therefore follows closely the text of the oral presentation at the Congress in 2007. Digital epigraphy has expanded rapidly in the years since then, and were I to survey the territory today, I might emphasize even more strongly than I did the importance of the free exchange of information within open communities of researchers working toward common goals. I have nonetheless refrained from modifying the text except to update references to the projects mentioned and to provide in the bibliography an expanded, if nonetheless selective, indication of the rich variety of research in digital epigraphy recently produced or currently under way.

[2] Panciera 2006a: 1913–14, from the introduction to a conference, *Dall'Orto Lapidario a Internet (Aquileia-Trieste, 14–16 novembre 2003)*.

Proceedings of the British Academy **177**, 275–96. © The British Academy 2012.

revolutionary, and the accomplishment of the great majority of it within the span of a few decades is indeed a wonder.

Boeckh's Greek corpus, first out of the gate, wrapped up its work in three folio volumes, which were so badly out of date by the time a fourth volume (of indices) was added in 1877 that the Berlin Academy had by then already decided to begin again *ab initio* with a successor, a work that ultimately became the *Inscriptiones Graecae*. Mommsen's corpus of Latin inscriptions, on the other hand, has endured and thrived, in part by keeping abreast of new developments not only in editorial techniques but also in technology. Since the publication in 1996 of a supplementary volume of the imperial inscriptions of Rome, texts have been transcribed in minuscule, with supplements and abbreviations expanded. They are equipped not only with drawn reconstructions (when needed and possible) but also with inset photographs that are more than adequate for most purposes. Ever more elaborate subject indices are being produced for various volumes of *CIL*, which open up the vast riches of the corpus to researchers with more particular interests and from more distant fields.[3] The pace of production can seem slow, but the editorial standards are the highest, and *L'Année épigraphique* keeps us admirably informed year to year.

Legitimate concerns are rightly raised about the longevity and accessibility of any of the digital storage media currently employed, whereas the codex has provided a remarkably versatile and sustainable means of conveying information for nearly two millennia. The internet was founded on the principle of redundancy, the efficacy of which has been demonstrated by the success of Gutenberg's technology, but no one who has stared in frustration and despair at an inert hard-drive when the processor refuses to work will doubt that information without retrieval is useless. A popular video depicting a medieval reader first learning how to 'access' a codex is funny precisely because the idea of a book being 'broken' or difficult to use seems to us absurd.[4] Systemic failures are said to be impossible—until they happen. Or the systems are superseded and replaced. There are no guarantees. The question, then, may reasonably be asked: is technology necessary for epigraphy? Or, perhaps more poignantly, is it worth it?

It may seem disingenuous to pose the question at such a late date. The IT revolution came to epigraphy in the latter half of the 1960s, at the University of Western Australia in Perth, where John Jory and a team of collaborators painstakingly keystroked into a database the nearly forty thousand texts included in volume six of the *CIL* pertaining to Rome, with the aim of producing a Key Word in Context

[3] For a brief history of the *Corpus Inscriptionum Latinarum*, a summary of its current state, and an outline of its projected plans through 2030, see Schmidt 2007: 8–43.

[4] 'Medieval Help Desk', produced in 2007 by the Norwegian Broadcasting Corporation (NRK), is most easily accessible on YouTube. For a version with subtitles in English, see <http://www.youtube.com/watch?v=pQHX-SjgQvQ&feature=related>.

(KWIC) index. Already then the data entry included a simple form of encoding, and since Jory was chiefly interested in establishing dating criteria based on a typology of formulae and letter forms, the markup focused on palaeographical features such as ligatures and tall letters. The decision not to encode irregular letter forms, such as tall Ts, that did not change the morphology of a word would not today be considered best practice, but one can only admire and applaud the initiative and energy that brought a path-breaking project to completion within seven years: researchers today still must negotiate compromises between richly encoded markup and timely results. The production standards of Jory's index were crude—the text was published directly from a computer print-out and the format was cumbersome: in bulk the mammoth six-tome set produced in 1974 and 1975, comprising more than seven thousand pages, weighed in at some fifty-five kilograms and occupied half as much shelf space again as the original text fascicles of *CIL* VI.[5]

It is difficult to understand, in retrospect, why the benefits of this valuable new tool were not welcomed and exploited more enthusiastically at the time. Perhaps the contrast between the crisp elegant typography and clean presentation of the original editions in the text volumes and the redeployment of the same words in messy, muddy block capitals interspersed with unfamiliar *sigla* discouraged readers from exploring and exploiting the new resource in ways we now consider standard. Another pioneer in digital epigraphy, David Packard, had anticipated this difficulty with computer print-outs and had already taken steps to alleviate it in producing a machine-generated concordance. It is bracing to read, in the preface to Packard's *Concordance to Livy*, published by Harvard University Press in 1968, not only the patient explanation of how to use a Key Word in Context Index and what possible purpose it might serve, but also why Packard in 1967 went to the trouble of writing programs not only for editing and concording the data but for operating a photo-typesetting machine in order to prepare camera-ready copy for the printer—both practices we today regard as commonplace.[6] Packard recognized already then that if the new tool (and the technology behind it) were to win acceptance, packaging would be important. The final product was made to look as similar as possible to the source from which it derived—the printed texts of the Oxford and Teubner editions—and was compact enough to fit into four manageable volumes. As a result the Livy concordance was hailed by literary scholars as a milestone and soon began to be used for analytical studies of the historian's text.

[5] E. J. Jory in Jory and Moore 1974–75, i–xx. Hassall 1977, 210 remarks the bulk of the volumes and laments the decisions not to encode materials and find-spots and to omit the *instrumentum domesticum* of Rome published in *CIL* XV. Jory's interest in palaeographical criteria for dating explains the curious attention given to the three Claudian letters, only two of which have thus far been found in Latin inscriptions: Oliver 1949.

[6] Packard 1968: I.v–vii marvels that the entire concordance, comprising some seventy-five million letters and stored on eleven miles of magnetic tape processed from sixty-five thousand keycards, was printed out in about three hours.

By the time Jory's *index verborum* appeared in 1974 and 1975, Packard was already at work in Palo Alto devising a system for transcribing polytonic Greek in standard ASCI characters and building a new dedicated computer (the Ibycus) designed to read and sort the electronic files of Greek texts that were being compiled by the *Thesaurus Linguae Graecae* (*TLG*). Packard's markup system, known as Beta Code, was officially adopted by the *TLG* in 1981 and quickly established itself as the standard means of encoding polytonic Greek. The value of the data being processed at *TLG* ensured that Beta Code held its dominant position for the next two decades, thus spurring the development on both sides of the Atlantic Ocean of software programs specifically designed to read it. The Packard Humanities Institute, founded in 1987, had by the end of that year released the first of its now famous CD-ROMs—the PHI disks—which included early versions of the Greek inscriptions then being compiled by the Cornell Epigraphy Project, along with a number of Latin literary texts. Five years later Jory produced a searchable version of his computerized concordance to *CIL* VI as a CD-ROM and packaged it together with a dedicated search engine designed to sort the material either in sanitized form, with encoding hidden from view, or as raw data 'with symbols for epigraphical details', such as tall letters.[7]

Meanwhile in Europe a number of researchers at different universities had launched various projects to encode and present ancient Greek and (especially) Latin inscribed texts in various ways—as integrated digital editions, as tagged entries in databases, or as raw data. These early initiatives were in many cases important for raising general awareness of the potential of computers to redeploy text in multiple ways, but they operated autonomously on individual platforms that were often incompatible with one another, and there was little coordination of effort. At the end of the 1980s an international colloquium on epigraphy and information technology held at Lausanne surveyed a number of these pioneering initiatives—in Europe at Bordeaux, Graz, Freiburg, Heidelberg, and Rome, and in North America, at Ithaca (New York) and Columbus (Ohio), as well as Palo Alto—and hammered out an agreement on what a basic epigraphic database should look like and what sort of tools would be needed to build and mine it.[8]

At that time computers and silicon chips had not yet infiltrated our lives as pervasively as they have done today. To remind us of how far we have come, Figure 13.1 shows an advertisement from that period for the first portable computer introduced by Apple in September of 1989, a few months after the colloquium at Lausanne was convened.[9]

[7] Jory 1992: CD-ROM brochure (1994).

[8] *Actes du Colloque Épigraphie et Informatique* 1989.

[9] Weighing more than seven kilograms and offered in the USA at a price of $7,300, the Apple Macintosh Portable boasted a hard drive storage capacity of forty megabytes, a processor speed of sixteen megahertz, a ten-inch black and white LCD screen, and a built in trackball to guide the cursor.

Figure 13.1. Apple Macintosh Portable computer, introduced September 1989

We had not yet then grown accustomed to systems of encoding and storing information being heralded one year as the new universal standard and swelling the rubbish heap of discarded technologies and outmoded practices a few years later. Most of us are familiar with changes in storage media—from floppy disks of two sizes, to Zip disks, to CDs, to USB memory sticks—but the more insidious shifts have occurred in the underlying methods of data encoding. At best they render obsolete markup superfluous, but more often they make it obstructive and sometimes impenetrable, thus vitiating much expended labour and costing more in addition. The information highway is littered with the wreckage of well-intentioned projects, including some of those pioneering ventures in epigraphic computing mentioned earlier that were unable to negotiate successfully an abrupt change in the course of the road. As a result of these two developments—the creeping advance of information technology into all aspects of our lives and the cooling of once fervent enthusiasms among converts chastened by the derailing of some brave early efforts—many now approach the question of technology in epigraphy with a mixture of resignation and wariness, if not outright scepticism.

The questions, then—is technology necessary for epigraphy, and is it worth it—are perhaps not entirely irrelevant even today. Indeed, the second should be asked by anyone venturing to design new ways to program computers to perform old tasks, or even new ones. To the first, is it necessary, we may simply note that since digital technologies are inevitable, they must at least be accommodated. If we wish to avoid a reputation within the scholarly community similar to that of the early sceptics of automobiles who stood by the road yelling 'Get a horse' as the future roared by, we must design new vehicles for the information highway. More positively, we may note that the trend in current scholarly editions of Greek and Latin inscriptions is for ever greater contextualization, not only of the text but of its support. In this, editing standards reflect the state of the field more generally, which has grown from being primarily the study of the texts written on stone and other hard surfaces to being the study of inscriptions in context, that is, of the texts and their carriers (of all sorts) together as ensembles, and, increasingly, of the unified statements that those ensembles of text and object present in their own physical settings. Innovations introduced into the entries in *CIL* just over a decade ago that employ new technologies of image reproduction have taken the medium of the codex, with its unavoidable spatial constraints (information must be presented at a fixed point in a page and pages at fixed places in a book, and so on), nearly as far as it can go.[10]

Through the simple mechanism of hypertext linking, the World Wide Web allows the possibility of crudely 'contextualizing' virtually any aspect of an inscription by pointing to relevant information stored elsewhere on the web, or of linking to multiple images representing various views of the object. This familiar technology makes the internet a natural medium for supporting this current trend in epigraphical research, and so might well seem, if not necessary, at least highly desirable. But it is for good reason that 'too much information!' has become a popular cry in the American vernacular: we are all daily bombarded with an excess of data either unwanted or packaged in ways inimical to our desired use of it. Epigraphists, who have always understood that interpreting any inscription means knowing at least a little about a lot of things, face this problem in their research as well. Far more often than in the past, the challenge facing us is not to locate and access information but to sift and sort it. The obvious solution to this rich man's dilemma is the synthetic integration, rather than the mere accumulation, of data, but how that might best be achieved is far from clear.

To the question 'Is it worth it?', a more cautiously positive answer may perhaps be returned. The potential benefits to epigraphists of the new information technology are by now widely recognized, but as Professor Panciera noted in his opening remarks (ch.12 above), potential is not actuality, and when it comes to results, our track record, until recently, has given little reason for optimism. No one who has

[10] For recent advances in the presentation of entries in *CIL*, see Schmidt 2007, 16, 25–7.

ever attempted to encode even the smallest amount of data will underestimate the labour involved in preparing text to be 'read' electronically by machines. Yet, from the beginning, encoders have preached the wisdom of 'front-loading' work at the data entry level, so that a well-designed software program can perform thankless tasks at a later stage with precision, speed, and accuracy. Already forty years ago Packard sounded this note in the preface to his *Concordance to Livy,* and it continues to drive the sales pitches that are necessary to secure funding for digital humanities projects today. Until now, however, the most pressing needs have dictated a minimum of front-loading and a maximum of integration of data and information readily available elsewhere, even if in undigested form. That process will no doubt continue, to our benefit, but it is perhaps not too rash to suggest that the time when the advantages of further aggregating and redistributing the existing data instead of refining and enriching it may be coming to an end. We now have the tools needed to perform efficiently the tasks required to produce integrated relational databases that allow both simple and complex searching of words, and the more difficult human challenges of breaking inertia and corralling lone wolves seem largely to have been met. In concluding this first part of my paper, then, allow me to hazard an oversimplified characterization of three stages—or rather phases, since they are not so much chronologically consecutive as they are ontologically distinct—that we have encountered thus far in our digital revolution, and to conclude with a guardedly optimistic prediction for the future.

The first phase of our IT revolution, as I would identify it, focused on the redeployment into more convenient forms of information traditionally viewed as essential, or at least desirable, in any edition of an inscription—most obviously the words of its text and an indication (ideally a representation) of its appearance. The early Key Word in Context indexes of Jory and others and the more recent volumes of *CIL*, with inset photographs and drawn reconstructions, are good examples of this form. Computers and the programs they run are in these cases used to rearrange words or images in newly useful ways but not to effect any essential changes in the data. The process is like moving checkers around a checkerboard, but it does not turn checkers into chess pieces. And the results are presented in a traditional form, a printed book.

The second phase may be said to have begun in the late 1980s with the software embedded in Packard's Ibycus mini-computer and the production in 1992 by a research firm in Melbourne, Australia, of the Epigraph processing program, packaged together with the data from Jory's *index verborum* on a CD-ROM.[11] This phase can itself be seen to fall into two distinct but overlapping stages. The first is characterized by the development of ad hoc search engines designed for individual platforms and a minimal encoding of data with very basic peripheral

[11] Jory 1992. The copyright for the EPIGRAPH software is held by Space-Time Research Pty Ltd.

information (collectively known as metadata), such as the provenance of the inscription (as in the *TLG* files) or on what part of the object the text is inscribed (as in Jory's database). The second, which flourishes today, involves a more elaborate markup of metadata. This often includes more detailed information about the object and letter forms, circumstances of discovery and preservation, and even historical information deduced about the contents of the text, for example, that a particular person belongs to the senatorial order or that a proper name is the name of a Roman legion. Figure 13.2 shows the range of categories of metadata, arrayed as search fields, that have been encoded to be read by one of the more elaborate and sophisticated of the search engines available for Latin inscriptions, that of the *Epigraphische Datenbank Heidelberg* (*EDH*).

EDH further typifies a second feature of this second stage of the second phase—its deployment via the internet rather than on a fixed platform. That crucial migration from a closed to an open medium, the linchpin of subsequent advances and the key to our digital future, has scarcely begun to be exploited to its full potential—nor indeed can it be, since new tools are being developed all the time by the engineers at Google and elsewhere—but it also marks the outer limits of this phase, as I am somewhat arbitrarily defining it. That is, in phase two, as I envisage it, the markup is restricted to semantic elements of the texts and objects—this bit of code means it is a *cognomen,* that bit a location, this bit a date, and so on—but the semantically tagged elements are sorted by categories, often hierarchically arranged, that nonetheless remain independent of one another. Although they can be, and often are, strategically arranged to bring information of the same order into proximity with similar information so that it can be easily compared, as with the groupings of the *EDH* Complex Search Fields, they function autonomously.

The third phase, which I think we may be on the cusp of realizing, will involve the syntactic analysis as well as the semantic markup of inscribed texts. By this I mean that the information will not only be categorized but will also be in part analysed digitally, based not only upon its own native features but also on its position and shape in relation to other elements. Thus, for example, a software program might automatically determine not only how many letters are likely to be missing from the end of a line truncated at the edge of a broken stone and what combinations of letters are possible to complete a fragmentary word, but also which supplements are most probable in any particular combination of lexical, grammatical, spatial, or geographical contexts. This concept is widely employed in artificial intelligence, and although computers cannot truly 'read' natural (human) language, a recent experiment in digital decoding of indecipherable script on the Vindolanda leaves here at Oxford has made considerable strides toward converting that vision to reality.[12] The highly abbreviated formulae of inscribed Latin texts,

[12] See Terras and Robertson 2005; Terras 2006, 123–52; and below, n. 27.

the relatively circumscribed range of contexts in which particular strings of letters or words regularly occur, and the close relationship of certain types of text to certain types of writing and their natural supports (for example, epitaphs carved on stone *stelae*, brickstamps impressed into the wet clay of roof tiles, curses scratched into lead tablets) collectively reduce considerably the variables that make true 'reading' so difficult for machines. These characteristics make Latin inscriptions ideal targets for experimentation with the auto-generated recognition of natural language.

Epigraphists, of course, need no help from machines to produce hypothetical supplements for missing text. Assistance with complex searches of information not adequately parsed by the customary Boolean options, however, would expand considerably the range of questions we might consider along the road to interpretation, and one can imagine many other potential benefits as well. Collectively, to attempt a general response to the question posed for this session—how may emerging technologies change the way we write history?—if the advances of Mommsen's era gave us elaborated word indexes (essentially categorized word lists) which enabled us to recognize patterns and irregularities of usage and thus to write a new kind of history about the human agents who produced them, then the advances of this third age of digital epigraphy, if I may call it that, will give us categorized sequences of 'phrases'. These may be literally groups of words arrayed in a semantically coherent fashion, or (what is potentially more exciting) may be a combination of a verbal phrase with a variety of other, semantically independent, elements: a location, a date, a material, an object, and so on—virtually any aspect of what we now recognize as contributing to an inscription's 'message'. Computers will seem to do some of the thinking for us, just as reasoned word indexes, properly compiled, seem to make certain deductions leap off the page, once a pattern of usage is recognized. In neither case does the tool perform or enable any new cognitive task. Rather, it makes possible the more sophisticated application of basic human cognition and judgement to more complex questions that have (in many cases) first been revealed by the process of pre-sorting that indexes do for words and that computers can do for more complex and non verbal signifiers. The two fundamental and necessary pre-conditions for the full implementation of this phase are the creation of a corpus of true electronic editions of inscriptions—something that the more elaborate digital databases already approach, although the data is not presented in conventional form—and the advent of the second generation of the World Wide Web, long heralded by its founding father and current director of the World Wide Web Consortium, Tim Berners-Lee, on which more below.

Time constraints prevent me from mentioning, even in passing, all the worthy projects that have contributed to the rich second phase of our IT revolution, as I have described it, or even all those that are truly excellent. Rather I propose to mention briefly a very few important advances in digital epigraphy during the past decade in three broad areas—databases, images, and editing—that have characterized the period.

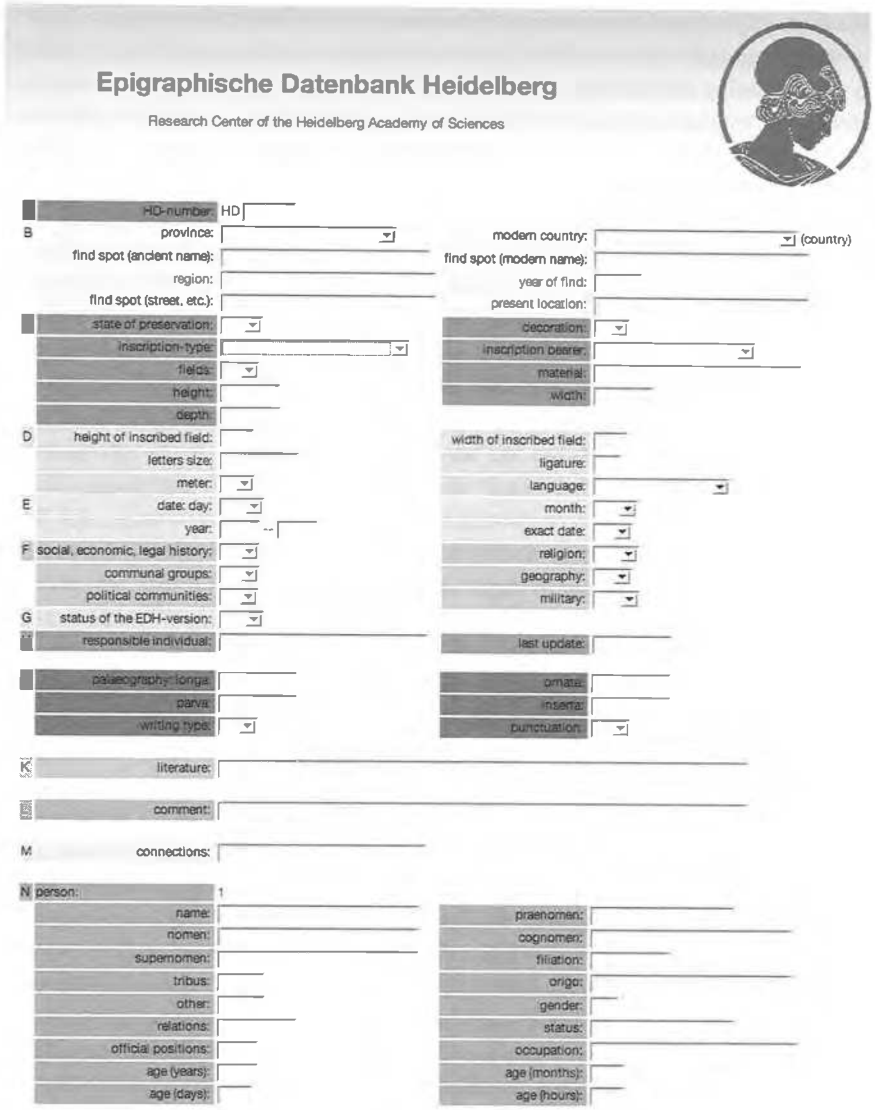

Figure 13.2. Complex search fields of the *Epigraphische Datenbank Heidelberg,* August 2007

I. Databases

I must here forego even a rapid summary of the vicissitudes and accomplishments of the two official AIEGL committees that have been charged, since 1989, with representing our organization's official role in the increasing spread of information technology throughout the field. The first was formed at the end of the colloquium in Lausanne; the second, which has guided us through a more fruitful but challenging period, was empanelled at the AIEGL Congress in Rome in 1997 and has been presided over since then by Silvio Panciera. The story can be traced in full through the Lausanne conference proceedings, periodic reports of the committees' work published in *Epigraphica*, and the texts of three communications by Panciera (two from conferences) now published in his *scritti vari*.[13] A crucial turning point came at a round-table meeting of the committee held at Aquileia and Trieste in November, 2003, when an original plan to create a unified database of all ancient Greek and Latin inscriptions was abandoned in favour of a more practical model: instead of creating an entirely new entity, the co-operating projects would join a federation of databases sharing a common portal but each retaining its autonomy and thus the possibility of developing, alongside the elements common to all the databases in the federation, tools particularly suited to its particular needs. Three projects centred on the Roman world that had already been co-ordinating their technical specifications formed the charter members of the new federation, which was to be headquartered at Rome under the acronym EAGLE—'Electronic Archive (at first, now Archives, in recognition of the independence of its federated members) of Greek and Latin Epigraphy'.[14] The decision to federate, based on the co-operative collaboration that had already been achieved among these three projects, at Heidelberg, Rome, and Bari, proved to mark a turning point. Since that time progress on all three has gathered momentum, and collectively, thanks to a co-ordinated and rational division of labour, they have substantially advanced our access to the Latin epigraphic patrimony of the Roman Empire. They have done so by making freely available on the World Wide Web the texts of nearly 100,000 Latin inscriptions and several hundreds of Greek texts and by enabling them to be searched in a uniform (though not yet unified) fashion. The federation is open, and recently a fourth important initiative, *Hispania Epigraphica*, accessible via the *Ubi Erat Lupa* website, has been welcomed into the community, and will soon add another 24,000 Latin texts to the pool.[15]

[13] See *Actes du Colloque Épigraphie et Informatique* 1989; Panciera 1999; 2004; 2005; 2006b. To these may now be added the report delivered by Panciera at the Oxford Congress (this volume).
[14] EAGLE: < http://www.eagle-eagle.it>.
[15] *Hispania Epigraphica* (*HE*) now has its own website, <http://www.eda-bea.es>, and *Ubi Erat Lupa* is now headquartered at the University of Salzburg, <http://www.ubi-erat-lupa.org/platform_e.shtml>. A committee on *Instrumentum* and IT charged by the AIEGL Board at a meeting in Ljubljana in June, 2005 with creating a unified database of inscribed *instrumentum* will eventually form another member of the EAGLE confederation.

The *Epigraphische Datenbank Heidelberg* (*EDH*), founded in 1986 and in many respects the grandfather of digital databases of Latin inscriptions, provided the essential framework for the structured databases of the EAGLE enterprise.[16] Guided for twenty years, from 1986 through 2006, by its founder Géza Alföldy, and since 2007 directed by Christian Witschel, *EDH*, under the EAGLE confederation, is responsible for encoding inscriptions from the Roman provinces of the empire. It has now made available online all three of its large databases, comprising some 56,000 texts, more than 14,000 photo images (another 10,000, archived as TIFF files, will eventually be added), and an Epigraphic Bibliography of some 11,000 items. Originally focused on inscriptions published since the completion of the first series of *CIL*, the researchers at *EDH* from the beginning have not merely transcribed the texts as published but have checked and re-edited them, so that the new presentations amount to new editions and are not simply electronic avatars of the texts printed in *L'Année épigraphique*, a fact not always properly recognized by those who cite them. With its own photographic holdings augmented by numerous web-links to other photographic archives, *EDH* is our single most useful electronic resource for basic research in Latin epigraphy and comes closest to providing, with its combined elements, the raw materials of an electronic edition of the epigraphic patrimony of the provinces of the Roman Empire.[17]

The *Epigraphic Database Roma* (*EDR*), under the directorship of Silvio Panciera and available online since 2004, is responsible for the inscriptions of Rome (excluding the Christian texts in *Inscriptiones Christianae Urbis Romae*, which are the provenance of a third member of the confederation), all of peninsular Italy as defined by its borders in Antiquity, Sicily, Sardinia, and the small coastal islands included in *CIL*.[18] The database is naturally focused on inscriptions discovered since the most recent of the series of supplementary volumes for Rome was published, that latter material having been surveyed already in Jory's printed index volumes and the *Epigraph* CD. However, plans are in place, with Jory's co-operation, to integrate the more than 39,000 texts of his database into the *EDR* material as well. More than 18,000 inscriptions are currently available to be searched at the *EDR* webpage in the same fashion as those at *EDH*.[19]

The third member of the confederation, the *Epigraphic Database Bari* (*EDB*), directed by Carlo Carletti and available on line since 2005, is responsible for the Christian inscriptions of Rome has already made available 26,000 of its nearly 35,000 texts, beginning with those from the suburban Christian cemeteries published in the new series of *ICUR*.[20]

[16] *Epigraphische Datenbank Heidelberg* (*EDH*): <http://www.epigraphische-datenbank-heidelberg.de>.
[17] See now Feraudi-Gruénais 2010b, 12–15; 2010c, 158; cf. Witschel 2010.
[18] *Epigraphic Database Roma* (*EDR*): <http://www.edr-edr.it>.
[19] For the history and structure of the project, see now Evangelisti 2010.
[20] *Epigraphic Database Bari* (*EDB*): <http://www.edb.uniba.it>. The full name of the project is *Epigraphic Database Bari. Documenti epigrafici romani di committenza cristiana, Secoli III–VIII.*

Regrettably, there are as yet too few examples of this sort to point to, but it is perhaps worth recalling that Jory's *index verborum* was not immediately hailed as the indispensable resource that we now recognize it to be, nor did even the superb indices of Dessau's *Inscriptiones Latinae Selectae* leave their mark on historical scholarship as soon as one would have supposed. In the modern era of instantaneous internet communication, we can and should do better. Thanks to the success thus far of the EAGLE initiative, the most important recent development in the IT revolution for Latin epigraphists, we now have the tools to interrogate the vast epigraphic corpus in increasingly sophisticated ways, performing in a matter of seconds searches that previously would scarcely have been undertaken because of the difficulty of sifting the potentially relevant data. [21] There is no longer any excuse not to do so, and although it may not yet be fair to say that the ways in which this capability can change how we write ancient history are limited only by our ingenuity, it is certainly true that we have not yet exploited the existing resources as effectively as we might. At the same time, the EAGLE enterprise is very much a work in progress, and others compiling or contemplating epigraphic databases of more limited scope should recognize that they will lose nothing in independence and will gain much in value and good will by contributing to the common pool.

II. Images

I have already mentioned the most important development in imaging, namely the advent of digital photography and the possibility that it provides, in conjunction with the internet, of disseminating high quality (if low resolution) images at relatively little cost. Adobe Photoshop can work wonders with images produced in poor light, but it is somewhat unsettling that we have not yet developed agreed standards for the responsible use of it. Epigraphists are well aware of the tricks that light and shadow can play on the eye with traditional film photography, but thus far there have been fewer notes of caution sounded about the wonders of digital imaging, nor has its potential to distort as well as to enhance been adequately explored. That said, several recent imaging projects have either avoided using light directly as a means of improving our ability to read previously illegible texts, or have employed it systematically to do more than to help us to see more clearly. I will mention three.

Perhaps the most exciting new technology for recovering inscribed texts is fluorescence imaging, a non-destructive technique whereby high energy x-ray

[21] An excellent illustration of how these new tools can illuminate our perspective was provided at the Oxford conference by the poster by Antonio Enrico Felle, based on research conducted through *EDB*, on the early Christian community represented in the Catacombs of Domitilla in Rome: Felle 2007.

beams are fired at the surface of an inscribed object, and the fluorescent photons refracted back from the face are read by a receiver to produce a chemical analysis of trace elements on the stone. Even when no longer visible to the eye, letter forms can thus be discerned by the remnants of iron and (for unknown reasons) lead, originally deposited by the tip of a sharpened chisel and subsequently sunk beneath the surface of the stone (Fig. 13.3).

Thus far the technique has been tested only on various limestones, but it has been successfully employed to identify the provenance of the black marble epitaph of the Carolingian Pope Hadrian I immured in the front of St Peter's basilica in Rome; to read two classical Attic and one Latin inscription in the collection of Columbia University; and, recently, to identify as a modern copy a replica

Figure 13.3. X-ray fluorescence imaging: a worn region of *CIL* VI 12139, from the centre of line 5, *A. Popil]ius A. l.[Chrestio*. From top to bottom: an optical image, an iron fluorescence image, and a lead fluorescence image (from Powers *et al.* 2005, 226, Figure 10).

of a fragment of the consular *fasti* from Teanum Sidicinum at New York University.[22]

A second type of procedure developed by engineers at the Hewlett Packard laboratories in Palo Alto, Polynomial Texture Mapping, uses the highlights created by light and shadow to extrapolate textures from a surface that help to eliminate the 'background noise' of non-significant marks.[23] Figure 13.4 shows an early example, employing a fixed light source on a Neo-Sumerian tablet, with the layers of texture superimposed systematically.

More recently, the technique has been improved by exploiting multiple lighting angles and a Java script extension to enable a viewer of the images on a web-browser to manipulate the direction of the 'light' by dragging the mouse, creating the effect of moving a light source such as a flashlight from side to side across the face of an inscription to provide raking shadows from multiple angles. In 2005 this newly enhanced version was exploited to help read inscribed writing on the ancient Greek astronomical computer found in a shipwreck dating from around 80 BCE and known as the Antikythera mechanism. [24] With a freely downloadable 'PTM Viewer' or Java extension made available by Hewlett-Packard, anyone with internet access and a home computer can now view pieces of the device in various exposures by moving a cursor around the screen.[25]

My third example, which I alluded to briefly earlier, employs imaging techniques in conjunction with a dialect of computer language used in artificial intelligence in order to decipher scripts predictively by analysing a known set of the script in

[22] For Pope Hadrian's epitaph, see Story et al. 2005. For the Columbia inscriptions, see Powers et al. 2005. For the fragment from Teanum Sidicinum, see now Powers et al. 2009. Widespread adoption of the technique seems unlikely, however, since it requires the use of a portable energy dispersive X-ray fluorescence (EDXRF) spectrometer (as was used on the Carolingian marble: Story et al. 2005: 168) or a high energy synchrotron (as at the Cornell High Energy Synchrotron Source (CHESS), where the other stones were examined). Robert Thorne, a physicist from CHESS, presented a report of the Cornell team's work at the Oxford Congress.

[23] The process was developed and originally described by Malzbender, Gelb, and Wolters (2001). A simple version of the technique known as 'shadow stereo' or 'phase congruency' has recently been successfully applied to the decipherment of a wooden stylus tablet from the Netherlands previously misread as describing the sale of an ox: Bowman, Tomlin, and Worp 2009: 158–9; Bowman, Crowther, Kirkham, and Prybus (2010), 92–6; below, n. 27.

[24] A regularly updated history of the discovery and ongoing study of the object is provided by the international Antikythera Mechanism Research Project at <http://www.antikythera-mechanism.gr/project>, with links to bibliography and downloadable images. In addition to Hewlett-Packard's Polynomial Texture Mapping, a powerful microfocus X-ray tomography machine developed by the British firm X-Tek in the same year produced high resolution digital radiographs of the mechanism that unexpectedly revealed on one fragment (G) more than 700 previously hidden inscribed characters: <http://www.xtekxray.com/applications/antikythera.html>.

[25] <http://www.hpl.hp.com/research/ptm/antikythera_mechanism/full_resolution_ptm.htm>. For other recent applications of Polynomial Texture Mapping in archaeology, see Earl, Martinez, and Malzbender 2010, esp. 6–7, on writing tablets from Herculaneum and Vindolanda, 7–9 on Roman brick stamps from Portus.

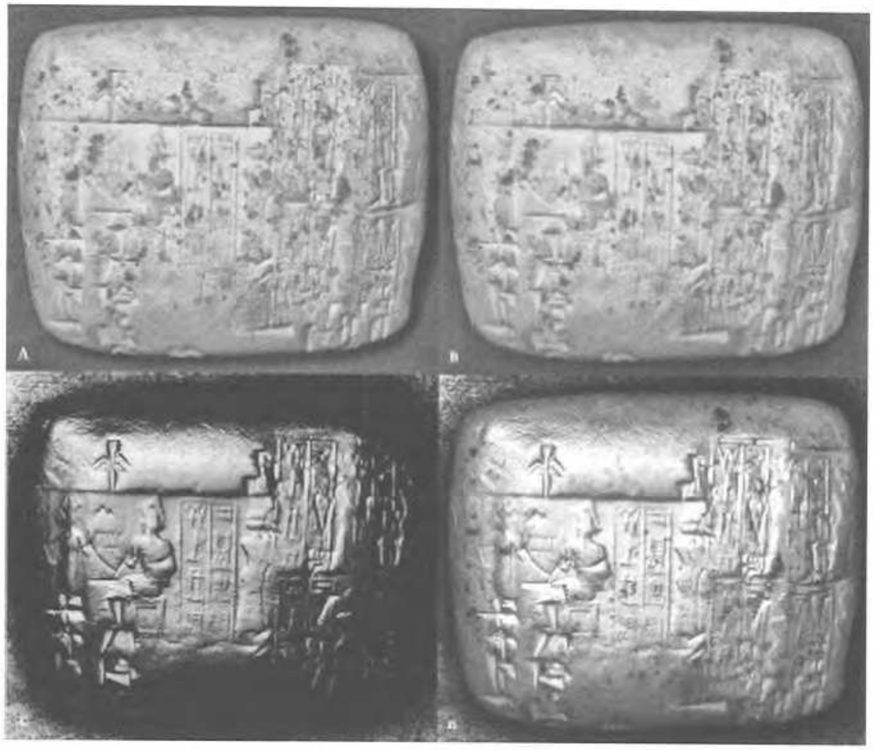

Figure 13.4. Four images of a neo-Sumerian tablet showing specular enhancement of an original photograph (A) by combining a reconstruction made from Polynomial Texture Mapping (B) with an image computed from extracted surface normals for individual pixels (C) to produce a composite image (D) (from Malzbender *et al.* 2001, 24, Figure 5)

order to project most likely outcomes of a range of possibilities of interpreting an indecipherable sample of it. The original system, known by its acronym GRAVA (for Grounded Reflective Adaptive Vision Architecture), was designed by Paul Robertson, a doctoral candidate of the Robots Research Group at Oxford in 2001, to 'read' a handwritten phrase by picking out in individual letters the junctures of strokes as well as their top and bottom points in order to create the letter schematically. This procedure will sound familiar to epigraphists acquainted with the method employed by Géza Alföldy and others to decipher monumental texts originally displayed in affixed bronze lettering by 'connecting the dots' of the nail holes left in the stone once the letters had been stripped away.[26] Another Oxford doctoral

[26] Alföldy's 'recovery' of the original dedication of the Colosseum provides a well-known illustration of the technique: see *CIL* VI 40454a.

candidate, Melissa Terras, working with Robertson and others at the Department of Engineering Science, has now adapted the original GRAVA system to assess the stroke position and angle rather than the end points of letters and has tested it experimentally on the written material from Vindolanda, both the inked wooden leaves and the tablets etched with a stylus, in order to understand better both how we 'read' scripts and how the scripts themselves might be predictively deciphered. The results of their exciting interdisciplinary research are reported in the monograph Terras recently published from her dissertation.[27]

III. Editing

Under this, the third and last of my categories, there is only one project to report: EpiDoc.[28] Ten years ago, a then doctoral candidate in ancient history at the University of North Carolina at Chapel Hill, Tom Elliott, began developing a system to encode the familiar Leiden editing conventions digitally so that they could be both read and interpreted by computers. Subsequently, with a small group of core collaborators, Elliott expanded the project into an attempt to create a fully operable system for editing ancient Greek and Latin inscriptions digitally, incorporating not only all the features we have come to expect in well-edited corpora but, as Professor Panciera has reminded us should be our ultimate goal, going beyond what we have traditionally been able to do to exploit the new medium to full advantage. At the foundation of the system was the conviction that digital epigraphy, in order to survive, would need to conform to the general standards being developed internationally for encoding digital information on the World Wide Web. Three issues in particular seemed crucial in this regard: choosing a computer language that would be adaptable to the needs of epigraphists and would not become obsolete as technology developed; employing a system of encoding that would conform to the way other languages are marked-up for electronic communication around the world; and finding a reliable and standardized way to write and display polytonic Greek characters digitally. The acronyms 'XML', for Extensible Markup Language (a sophisticated relative of the more familiar hypertext markup language, html) and 'TEI', for Text Encoding Initiative (a consortium of several hundreds of text encoding projects around the world that have agreed to follow the same guidelines), are perhaps less familiar than the third term, 'Unicode', but these three entities, all

[27] Terras 2006. Terras 2010, 180–2 reports on the continuation of research on digitally assisted reading of ancient documentary texts under the auspices of a new project, eSAD (for e-Science and Ancient Documents), closely linked with a related project at Oxford's Center for the Study of Ancient Documents, 'A Virtual Research Environment for the Study of Documents and Manuscripts', for which see Bowman, Crowther, Kirkham, and Prybus 2010.
[28] EpiDoc: <http://epidoc.sourceforge.net>.

now well established, have provided the answers to those challenges and have underpinned the development of EpiDoc from the beginning.[29]

Progress in the early stages of development was slow, but two pilot projects in particular have effectively demonstrated the possibilities of the new tool, The Vindolanda Tablets Online, and Inscriptions of Aphrodisias, both of which have produced dynamic digital editions that update and supersede original print publications.[30] Plans are now under way to convert the data already entered into the major EAGLE databases at Heidelberg and Rome into XML EpiDoc, so that the hard work of encoding already performed will not have to be redone. Recently the EpiDoc system has been adopted and its development encouraged by large papyrological projects in Germany and the United States, and by numismatics organizations in the United Kingdom and New York, and thus seems to be establishing itself as the standard method of encoding documentary texts and objects from Antiquity.[31]

IV. Conclusion

To conclude, I hope in the last paragraphs to have provided some indication of why I feel optimistic about the future of digital epigraphy. Momentum is with us. The most important recent advance—the establishment of the EAGLE confederation—and the major contributions its three current members have already made to our knowledge and ability to query our primary sources have already borne the

[29] For a basic introduction to XML, see <http://www.w3.org/XML/1999/XML-in-10-points>. For the Text Encoding Initiative (TEI), <http://www.tei-c.org/index.xml>. For Unicode, <http://unicode.org>. The interests of epigraphists in reproducing on the internet symbols for editing conventions and special characters found mainly in inscriptions are well served by the open-source Cardo font created by independent scholar, David Perry in 2002 and regularly updated: see <http://scholarsfonts.net/cardofnt.html>.

[30] *The Vindolanda Tablets Online* edition (2003–), based on the print publications of the first two volumes of the corpus of tablets, Bowman and Thomas 1983 and Bowman and Thomas 1994, revises and updates both: <http://vindolanda.csad.ox.ac.uk>. The *Inscriptions of Aphrodisias Project* in 2004 produced a revised edition of an original print publication, Roueché 1989, *ala2004* <http://insaph.kcl.ac.uk/ala2004>, and in 2007 introduced at the Oxford Congress a digital first edition of the inscriptions of Aphrodisias recorded up to 1994: Reynolds, Roueché, and Bodard 2007 (*IAph*), <http://insaph.kcl.ac.uk/iaph2007>.

[31] Further on the history of EpiDoc, now widely recognized as standard for the digital publication of ancient documents, and the development of a growing international EpiDoc community, see <http://epidoc.sourceforge.net/#about>; Bodard 2008 and 2010; and Tupman 2010, 78–81. For a view into the future of digital epigraphy and EpiDoc's place in it see Cayless, Roueché, Elliott, and Bodard (2009), esp. 1.3. The *Integrating Digital Papyrology (IDP) Project* in 2008 completed its conversion of the large databases of digital files of papyri at Duke and Heidelberg from SGML to XML Epidoc and from BetaCode to Unicode precisely in order to make them more easily integrated with other databases of ancient documents: see <http://idp.atlantides.org/trac/idp/wiki/BackgroundAndFunding>.

fruit of enlisting a fourth major participant. This is clearly the way forward, and (to repeat), anyone currently compiling or contemplating compiling a database of Greek or Latin inscriptions would be well advised to join this growing movement. After a slow start, the system of encoding text in TEI conformant XML developed by the EpiDoc community is likewise now rapidly gaining ground and is winning important new adherents not only among epigraphists but also among papyrologists and numismatists. Advanced imaging techniques are being employed not only to enhance our ability to read illegible texts but to decipher scripts and the ways they are read. Things are coming together.

Finally, a word on the infrastructure that I mentioned earlier as so crucial to our success or failure in all these areas. No sooner had Tim Berners-Lee and Robert Cailliau fathered the World Wide Web in 1990 than Berners-Lee began talking about and planning its successor, which he first articulated at a conference in 1994 and first presented formally as the Semantic Web in 2001.[32] Whereas the World Wide Web mainly links documents for humans to read, the Semantic Web would link data and information for computers to manipulate. Its underlying architecture would depend upon well-established Web standards for expressing shared meaning though a common language that enables interoperability between systems. Once realized, it promises to allow the sort of syntactically complex searching and sorting I alluded to earlier, and to enable much of it to happen automatically through the work of digital agents. Since 1999 Berners-Lee has been describing the basic structure of the Semantic Web as a 'layer cake' or a 'stack of expressive power' and representing it with a diagram such as that reproduced as Figure 13.5.

The most significant developments in the past decade have occurred within the middle registers, but for our purposes the lower layers are more important. At the bottom left of the stack lies the acronym URI, for Uniform Resource Indicator. This is more familiar to us from its enhanced counterpart, the Uniform Resource Locator (URL), which enables webpages on the World Wide Web to be found. At the right lies Unicode. Just above and occupying an entire level, XML undergirds the rest of the structure above. That the principles of digital editing in EpiDoc share this bedrock foundation of technology with the anticipated Semantic Web gives reason to believe that the future of digital epigraphy will not be left in the past. If we can provide the crucial human element needed for its ultimate success, a willingness to collaborate and a commitment to open sharing of information, we can share also the distinction of our forebears in digital epigraphy during the 1960s by contributing to as well as benefiting from the next wave of the IT revolution.

[32] Berners-Lee, Hendler, and Lassila 2001; <http://www.w3.org/2001/sw>. See now also Shadbolt, Hall, and Berners-Lee 2006 and Feigenbaum et al. 2007 for progress and remaining challenges.

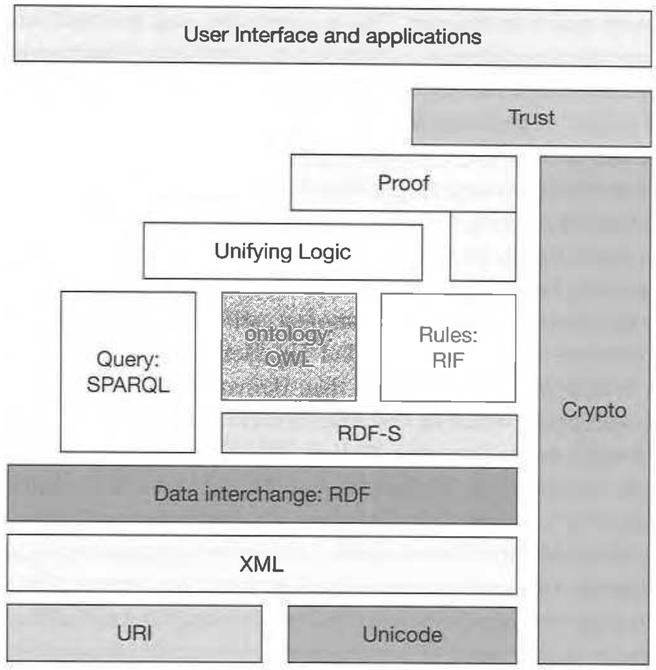

Figure 13.5. Semantic Web 'Stack of expressive power', from Berners-Lee 2007, slide 20

References

Actes du Colloque Épigraphie et Informatique (1989). *Actes du Colloque Épigraphie et Informatique: 26–27 mai 1989 Centre Audio-visuel de l'Université* (Lausanne).

Benefiel, R. R. (2010), 'Rome in Pompeii: Wall Inscriptions and GIS', in Feraudi-Gruénais (2010a), 45–75.

Berners-Lee, T. (2007), 'Introduction to the Semantic Web', slideshow, MIT Computer Science and Artificial Intelligence Laboratory, <http://dig.csail.mit.edu/2007/Talks/0108-sw-tbl/#%281%29>.

Berners-Lee, T., Hendler, J., and Lassila, O. (2001), 'The Semantic Web', *Scientific American* May 2001, 34–43.

Bodard, G. (2008), 'The *Inscriptions of Aphrodisias* as Electronic Publication: A User's Perspective and a Proposed Paradigm', *Digital Medievalist* 4, <http://www.digitalmedievalist.org/journal/4/bodard/>.

—— (2010), 'EpiDoc: Epigraphic Documents in XML for Publication and Interchange', in Feraudi-Gruénais (2010a), 101–18.

Bodard, G. and Mahony, S. (eds) (2010), *Digital Research in the Study of Classical Antiquity*, (Farnham and Burlington VT).

Bodard, G., Dunn, S., Garcés, J., Mahony, S. et al. (2006–). *The Digital Classicist*, <http://wiki.digitalclassicist.org/Main_Page>.

Bowman, A. and Thomas, D. (1983), *Vindolanda: The Latin Writing Tablets* (London).

Bowman, A. and Thomas, D. (1994), *The Vindolanda Writing-Tablets: (Tabulae Vindolandenses II)* (London).

Bowman, A. K., Crowther, C. V., Kirkham, R., and Prybus, J. (2010), 'A Virtual Research Environment for the Study of Documents and Manuscripts', in Bodard and Mahony (2010), 87–103.

Bowman, A. K., Crowther, C. V., Pearce, J., *The Vindolanda Tablets Online* (2003–), <http://vindolanda.csad.ox.ac.uk>.

Bowman, A. K., Tomlin, R. S. O., and Worp, K. A. (2009), '*Emptio Bovis Frisica*: the "Frisian Ox Sale" Reconsidered', *JRS* 99, 156–70.

Cayless, H., Roueché, C., Elliott, T., and Bodard, G. (2009), 'Epigraphy in 2017', in Crane and Terras (eds) (2009).

Crane, G. and Terras, M. (eds) (2009), *Changing the Center of Gravity: Transforming Classical Studies Through Cyberinfrastructure*, Digital Humanities Quarterly 3.1, <http://digitalhumanities.org/dhq/vol/3/1/000030/000030.html>.

Crane, G., Seales, B., and Terras, M. (2009), 'Cyberinfrastructure for Classical Philology' in Crane and Terras (eds) (2009).

Earl, G., Martinez, K., and Malzbender, T. (2010), 'Archaeological Applications of Polynomial Texture Mapping: Analysis, Conservation and Representation', *Journal of Archaeological Science* 37, 1–11.

Elliott, T. et al. (2007–), *Epidoc: Guidelines for Structured Markup of Epigraphic Texts in TEI XML* <http://epidoc.sourceforge.net/index.shtml>.

Evangelisti, S. (2010), '*EDR*: History, Purpose, and Structure', in Feraudi-Gruénais (2010a), 119–34.

Feigenbaum, L., Herman, I., Hongsermeier, T., Neumann, E., and Stephens, S. (2007), 'The Semantic Web in Action', *Scientific American* Dec., 90–7.

Felle, A. E. (2007), 'La documentazione epigrafica della catacomba di Domitilla a Roma alla luce dell'*Epigraphic Database Bari*: nuovi elementi di riflessione', <http://ciegl.classics.ox.ac.uk/html/posters.shtml>, no. 29.

Feraudi-Gruénais, F. (2010a), *Latin on Stone: Epigraphic Research and Electronic Archives* (Lanham, MD).

—— (2010b), 'Latin in Stone: Epigraphy and Databases', in Feraudi-Gruénais (2010a), 1–19.

—— (2010c), 'An Inventory of the Main Electronic Archives of Latin Inscriptions', in Feraudi-Gruénais (2010a), 157–60.

Hassall, M. W. C. (1977), Review of Jory and Moore 1974–75, *JRS* 67, 209–11.

Jory, E. J. (1992), *Epigraph. A Database of Roman Inscriptions*, CD-ROM (Melbourne).

Jory, E. J. and Moore, D. W. (1974–75), *Corpus Inscriptionum Latinarum* vol. VI, pars vii, *Indices Vocabulorum*, fascs. 1–6 (Berlin).

Malzbender, T., Gelb, D., Wolters, H. (2001), 'Polynomial Texture Maps', Hewlett-Packard Laboratories, in SIGGRAPH '01 (Proceedings), 519–28. <http://www.hpl.hp.com/ptm>.

Oliver, R. P. (1949), 'The Claudian letter I-', *AJA* 53, 249-57

Packard, D. W. (1968), *A Concordance to Livy, Volumes I-IV* (Cambridge MA).

Panciera, S. (1999), 'Nouvelles de l'A.I.E.G.L. 1999', *Epigraphica* 61, 311–13.

—— (2004), 'Nouvelles de l'A.I.E.G.L. 2004: Commission Epigraphie et Informatique', *Epigraphica* 66, 392–3.

—— (2005), 'Commission Informatique et Instrumentum', in 'Nouvelles de l'A.I.E.G.L. 2005', *Epigraphica* 67, 559.

—— (2006a), *Epigrafi, epigrafia, epigrafisti. Scritti vari editi e inediti (1956–2005) con note complementari e indici*, 3 vols., Vetera 16 (Rome).

—— (2006b), 'EAGLE: Cronostoria di un problema e di un progetto', in Panciera (2006a), 1913–17.

Powers, J., Dimitrova, N., Huang, R., Smilgies, D.-M., Bilderback, D. H., Clinton, K., and Thorne, R. E. (2005), 'X-ray Fluorescence Recovers Writing from Ancient Inscriptions', *ZPE* 152, 221–7.

Powers, J., Smilgies, D.-M., Geil, E. C., Clinton, K., Dimitrova, N., Peachin, M., Thorne, R. E. (2009), 'X-ray Fluorescence Imaging Analysis of Inscription Provenance', *Journal of Archaeological Science* 36.2, 343–50.

Reynolds, J., Roueché, C., and Bodard, G. (2007), *Inscriptions of Aphrodisias (IAph)* <http://insaph.kcl.ac.uk.iaph2007/>.

Roueché, C. (1989), *Aphrodisias in Late Antiquity: The Late Roman & Byzantine Inscriptions including Texts, from the Excavations at Aphrodisias Conducted by Kenan T. Erim*, with contributions by J. M. Reynolds, *JRS* monographs 5 (London).

Roueché, C., Reynolds, J., and Bodard, G. (2004), *Aphrodisias in Late Antiquity: The Late Roman and Byzantine Inscriptions*, revised second edition (*ALA2004*) <http://insaph.kcl.ac.uk/ala2004/>

Shadbolt, N., Hall, W., and Berners-Lee, T. (2006), 'The Semantic Web Revisited', *IEEE Intelligent Systems* May/June 2006, 96–101 <http://eprints.ecs.soton.ac.uk/12614/1/Semantic_Web_Revisted.pdf>.

Schmidt, M. (2007), *Corpus Inscriptionum Latinarum*, with English translation by O. Mulholland (Berlin).

Story, J., Bunbury, J., Felici, A. C., Fronterotta, G., Piacentini, M., Nicolais, Ch., Scacciatelli, D., Sciuti, S., and Venditelli, M. (2005), 'Charlemagne's Black Marble: The Origin of the Epitaph of Pope Hadrian I', *PBSR* 73, 157–90.

Terras, M. M. (2006), *Intelligent Systems to Aid Historians in the Reading of the Vindolanda Texts* (Oxford).

Terras, M. (2010), 'The Digital Classicist: Disciplinary Focus and Interdisciplinary Vision', in Bodard and Mahony (2010), 171–89.

Terras, M. and Robertson, P. (2005), 'Image and Interpretation: Using Artificial Intelligence to Read Ancient Roman Texts', *Human IT* 7.3, 1–56 <http://www.hb.se./bhs/ith/3–7/mtpr.pdf>.

Tupman, C. (2010), 'Contextual Epigraphy and XML: Digital Publication and its Application to the Study of Inscribed Funerary Monuments', in Bodard and Mahony (2010), 73–86.

Witschel, C. (2010), 'The Epigraphic Habit in Late Antiquity: An Electronic Archive of Late Roman Inscriptions Ready for *Open Access*' in Feraudi-Gruénais (2010a), 77–97.